

# Quantum Advantage Lab

An Interactive Platform for Visualizing Quantum vs. Classical Algorithm Races

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## 1. Concept Description

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Quantum Advantage Lab is an interactive web platform that races quantum algorithms against their classical counterparts in real time, making quantum speedups visible and intuitive. Users launch side-by-side comparisons and watch animated D3.js visualizations as both solvers progress toward a solution simultaneously.

The platform ships with four flagship modules:

- **Search:** Grover's algorithm vs. brute-force linear search on unstructured databases.
- **Graph Traversal:** Quantum walks vs. classical random walks on complex graphs.
- **Optimization:** Variational Quantum Eigensolver (VQE) vs. classical gradient-based optimization for molecular ground-state energy.
- **Simulation:** Hamiltonian simulation on quantum hardware vs. classical matrix exponentiation.

Each module executes real quantum circuits on IonQ trapped-ion hardware via Qiskit, exploiting IonQ's all-to-all qubit connectivity and high gate fidelities to keep circuits shallow enough that quantum advantage is visible, not buried in noise. Intermediate results stream to the browser so users see the quantum computation unfold — not just the final answer. Step-by-step annotations explain what is happening at each stage, bridging the gap between abstract theory and tangible experience. The entire platform is open-source (MIT license), designed for remixing and community extension.

## 2. Technical Approach

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### Architecture Overview

**Frontend:** React + Vite | **Visualization:** D3.js | **Backend:** FastAPI + WebSocket | **Quantum:** Qiskit + IonQ Provider

### Why IonQ's Trapped-Ion Architecture

This project is designed around IonQ's trapped-ion QPU — not as a generic backend, but because its architecture uniquely enables every module:

- **All-to-all connectivity:** IonQ's ions can entangle with any other ion directly, without the SWAP-gate overhead that superconducting architectures require for non-adjacent qubits. For Grover's multi-qubit oracle and VQE entangling layers, this means dramatically shallower circuits — fewer gates, less accumulated error, and a cleaner signal for the visualization to display.
- **High-fidelity gates:** IonQ Aria-class systems achieve >99.5% two-qubit (MS gate) fidelity. This is critical for our "race" concept: users need to see quantum algorithms actually win, not drown in noise. High fidelity lets us run deeper Grover iterations and more VQE optimization rounds before decoherence degrades results.
- **Long coherence times:** Trapped-ion qubits maintain coherence for seconds (vs. microseconds for superconducting qubits), enabling the multi-step Hamiltonian simulation and iterative VQE circuits that would decohere on other platforms at these qubit counts.
- **Native gate set (GPI, GPI2, MS):** We compile directly to IonQ's native gates rather than using generic transpilation, minimizing circuit depth. The Mølmer-Sørensen (MS) entangling gate is particularly efficient for the all-pairs interactions in quantum walk and simulation modules.

In short, IonQ's hardware gives us the fidelity to show real quantum advantage at educational scale, and the connectivity to keep circuits compact enough that advantage is visible rather than buried in routing overhead. A local Aer simulator provides an instant-feedback "practice mode" for exploration without consuming hardware credits.

## Quantum Circuit Layer

All circuits are built with **Qiskit** and dispatched via the Qiskit IonQ provider, compiled to IonQ's native {GPI, GPI2, MS} gate set. Each module defines a parameterized circuit template: Grover oracles are auto-generated from the search target and leverage all-to-all connectivity for single-depth oracle construction; VQE ansatzes use hardware-efficient layouts that exploit full connectivity rather than linear-chain entanglement; and Hamiltonian simulation circuits employ first-order Trotterization with native MS gates for efficient pairwise interactions.

## Real-Time Streaming Backend

A FastAPI server manages job orchestration. When a user starts a race, the backend submits the quantum job to IonQ and launches the classical solver concurrently. A WebSocket connection streams incremental state to the browser: partial search progress, intermediate VQE energy estimates, or random-walk positions. This architecture keeps the frontend reactive and the experience compelling even when quantum jobs take seconds to return.

## Visualization Frontend

The React frontend renders two synchronized D3.js panels — one for each solver. Animations include search-space heatmaps (Grover's), force-directed graph traversals (walks), energy-landscape contour plots (VQE), and matrix-norm decay curves (simulation). A shared timeline scrubber lets users replay any race at their own pace.

## Modular Design

Each comparison module is a self-contained package: a circuit builder, a classical solver, a visualization component, and an explanatory narrative. This plug-in architecture lets the community contribute new modules (e.g., Shor's algorithm, QAOA for MaxCut) without modifying core infrastructure.

## 3. Creative and Educational Value

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Most introductions to quantum computing describe speedups with asymptotic notation and circuit diagrams. Quantum Advantage Lab takes a fundamentally different approach: it lets users watch quantum advantage happen. Seeing Grover's algorithm collapse toward the solution in  $\sqrt{N}$  steps while brute-force plods through  $N$  entries makes the quadratic speedup visceral, not theoretical.

Every module includes layered annotations — a plain-language summary for newcomers, plus expandable technical detail (gate decompositions, circuit depth, error rates) for advanced learners. This makes the platform useful across skill levels, from a curious teenager's first encounter with quantum computing to a graduate seminar's hands-on lab.

Educationally, Quantum Advantage Lab doubles as courseware. Instructors can assign specific modules, adjust problem sizes, and have students compare simulator results to real hardware outputs — introducing noise, decoherence, and error mitigation organically. The MIT license and modular architecture encourage community remixes: hackathon teams can build new modules, and researchers can plug in custom circuits to visualize their own algorithms. The goal is a living, growing resource that makes quantum computing tangible, shareable, and genuinely fun.

## 4. Budget and Compute Estimates

### Development Costs

| Item                                                     | Cost           |
|----------------------------------------------------------|----------------|
| Frontend development (React, D3.js visualizations)       | \$2,000        |
| Backend & WebSocket infrastructure (FastAPI, deployment) | \$1,500        |
| Documentation, testing, and CI/CD                        | \$1,000        |
| Hosting (6 months, cloud compute)                        | \$500          |
| <b>Development Subtotal</b>                              | <b>\$5,000</b> |

### IonQ Compute Credits

| Module                                    | Circuits / Run | Est. Runs | Cost Estimate    |
|-------------------------------------------|----------------|-----------|------------------|
| Grover's Search (up to 20 qubits)         | 50-200         | 5,000     | ~\$12,000        |
| Quantum Walks (8-15 qubits)               | 20-80          | 4,000     | ~\$10,000        |
| VQE Optimization (6-12 qubits, iterative) | 200-500        | 3,000     | ~\$15,000        |
| Hamiltonian Simulation (10-20 qubits)     | 30-100         | 3,000     | ~\$13,000        |
| <b>Compute Subtotal</b>                   |                |           | <b>~\$50,000</b> |

**Total Requested Budget: ~\$55,000** (\$5K development + ~\$50K IonQ credits)

A local simulator mode ensures the platform remains fully functional for exploration and teaching without consuming hardware credits.

## 5. Timeline and Deliverables

| Week | Dates | Milestone | Deliverable |
|------|-------|-----------|-------------|
|------|-------|-----------|-------------|

|   |                |                                  |                                                                            |
|---|----------------|----------------------------------|----------------------------------------------------------------------------|
| 1 | Apr 13 – 19    | Core infrastructure              | FastAPI backend, WebSocket streaming, React shell, CI/CD pipeline          |
| 2 | Apr 20 – 26    | Module 1: Grover's Search        | Search race with animated heatmap visualization, IonQ integration tested   |
| 3 | Apr 27 – May 3 | Module 2: Quantum Walks          | Graph traversal race with force-directed animation                         |
| 4 | May 4 – 10     | Module 3: VQE Optimization       | Energy-landscape race with contour-plot visualization                      |
| 5 | May 11 – 17    | Module 4: Hamiltonian Simulation | Simulation race with matrix-norm decay curves                              |
| 6 | May 18 – 25    | Polish & launch                  | UX refinement, documentation, community contribution guide, public release |

## 6. Team

**Hossein Sadeghi Esfahani, PhD** — Sole proposer and developer.

Hossein holds a PhD from the University of British Columbia and brings over a decade of experience at the frontier of quantum computing. As former Director of Partnerships and Business Development at Pasqal Canada, he led quantum computing go-to-market strategy across North America. Prior to that, he spent seven years at D-Wave Systems as a Principal Technical Business Development Manager, where he co-invented three patents in quantum optimization and benchmarking and led engagements with Fortune 500 clients deploying quantum solutions in production.

His research contributions include publications in quantum machine learning and optimization. He has served as an investigator in the Creative Destruction Lab (CDL) Quantum Stream, mentoring early-stage quantum startups. Hossein combines deep technical fluency in quantum algorithms with extensive experience translating complex technology into accessible, high-impact products — the exact skill set this project demands.

